GEOG60941
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A multi-temporal Landsat NDVI analysis (2000–2024) of vegetation disturbance and recovery on steep Himalayan terrain in Himachal Pradesh, India.

1. Introduction

Mountain ecosystems are very prone to disturbances, mainly due to their steepness, thin soils, and the close relationships among plants, water, and the stability of slopes. In the western Himalayas, landslide hazards have become a major issue due to the increasing climatic variability and human pressure. Vegetation is the first line of defense against landslides as it holds on to the soil and controls the runoff; its disappearance or slow return can make the area more vulnerable to shallow landslides, especially in Himachal Pradesh (Chen et al., 2019; Pacheco-Quevedo et al., 2023).

Event-based landslide inventories are typically spatially incomplete and fail to reflect gradual or cumulative patterns of environmental degradation. Field-based monitoring faces similar limitations due to the challenging accessibility of Himalayan terrain. Earth Observation (EO) time-series analysis offers a spatially consistent means of assessing vegetation disturbance, recovery, and decline over large areas. Satellite-derived NDVI is widely used to quantify vegetation condition, with temporal trajectories revealing degradation not evident from single-date imagery (Forkel et al., 2013; Roy et al., 2014).

The relevance of vegetation disturbance depends strongly on terrain properties such as slope and elevation. Regional studies have identified forest degradation and slow recovery as key conditioning factors (Gautam et al., 2022; Singh et al., 2024), yet most rely on short temporal windows. This study applies a Landsat-based NDVI and LandTrendr framework (2000–2024) to link vegetation disturbance, recovery duration, and long-term degradation with steep terrain (≥30°), enabling assessment of landscape-scale environmental stress relevant to landslide monitoring.

This study aims to quantify and spatially characterise long-term vegetation disturbance and recovery on steep Himalayan terrain, and to identify zones experiencing persistent landslide-relevant environmental stress. This aim is addressed through the following research questions:

• How has vegetation disturbance expanded across Himachal Pradesh (2000–2024), and what proportion occurs on slopes ≥30°?
• Are disturbance and delayed recovery preferentially concentrated on steep slopes and specific elevation bands?
• Do disturbed steep-slope areas exhibit stronger negative NDVI Sen-slope trends than nearby stable terrain?
• Are disturbance magnitude, recovery duration, and NDVI degradation stratified by slope and elevation class?
• Which districts show the strongest combined disturbance, recovery delay, and NDVI degradation stress signals?

2. Methods

This study applies a multi-temporal remote sensing framework to identify, characterise, and spatially contextualise vegetation disturbance and recovery across the steep Himalayan terrain of Himachal Pradesh between 2000 and 2024. Himachal Pradesh, located in the north-western Himalaya of India, was selected as the study area due to its pronounced topographic heterogeneity and extensive steep terrain, which create strong gradients in climate, vegetation, and geomorphic processes relevant to slope instability. The state spans elevations from low foothills to high alpine zones and is characterised by rugged mountain landscapes, forest-dominated slopes, and increasing anthropogenic pressure. All analyses were spatially constrained to the official state boundary of Himachal Pradesh (Fig. 1).The framework is explicitly designed to monitor landscape-scale environmental stress, rather than predict individual landslide events, aligning with perspectives that conceptualise landslides as outcomes of interacting environmental stressors (Chen et al., 2019; Pacheco-Quevedo et al., 2023).

Fig. 1. Himachal Pradesh Study are and Elevation Bands

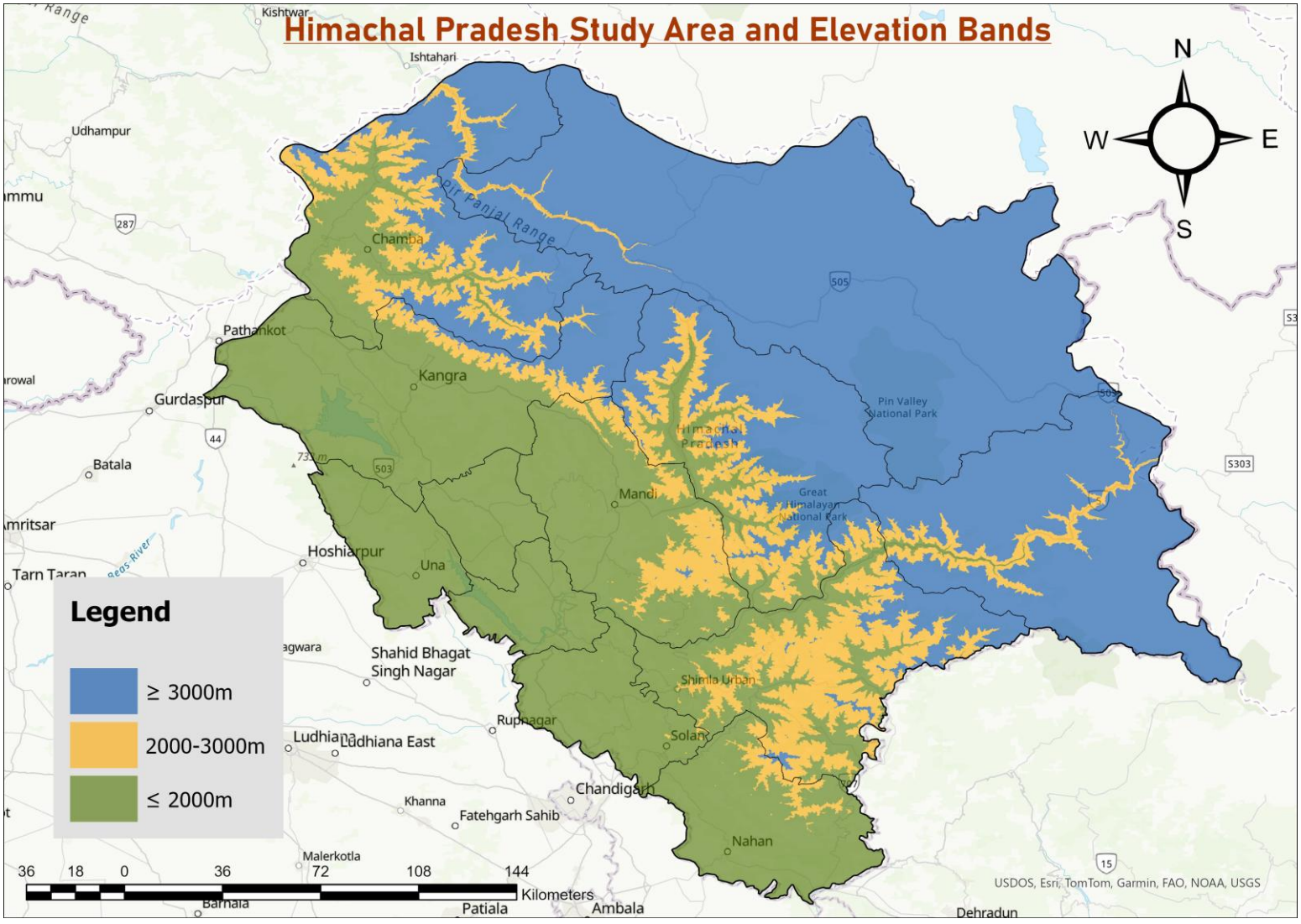
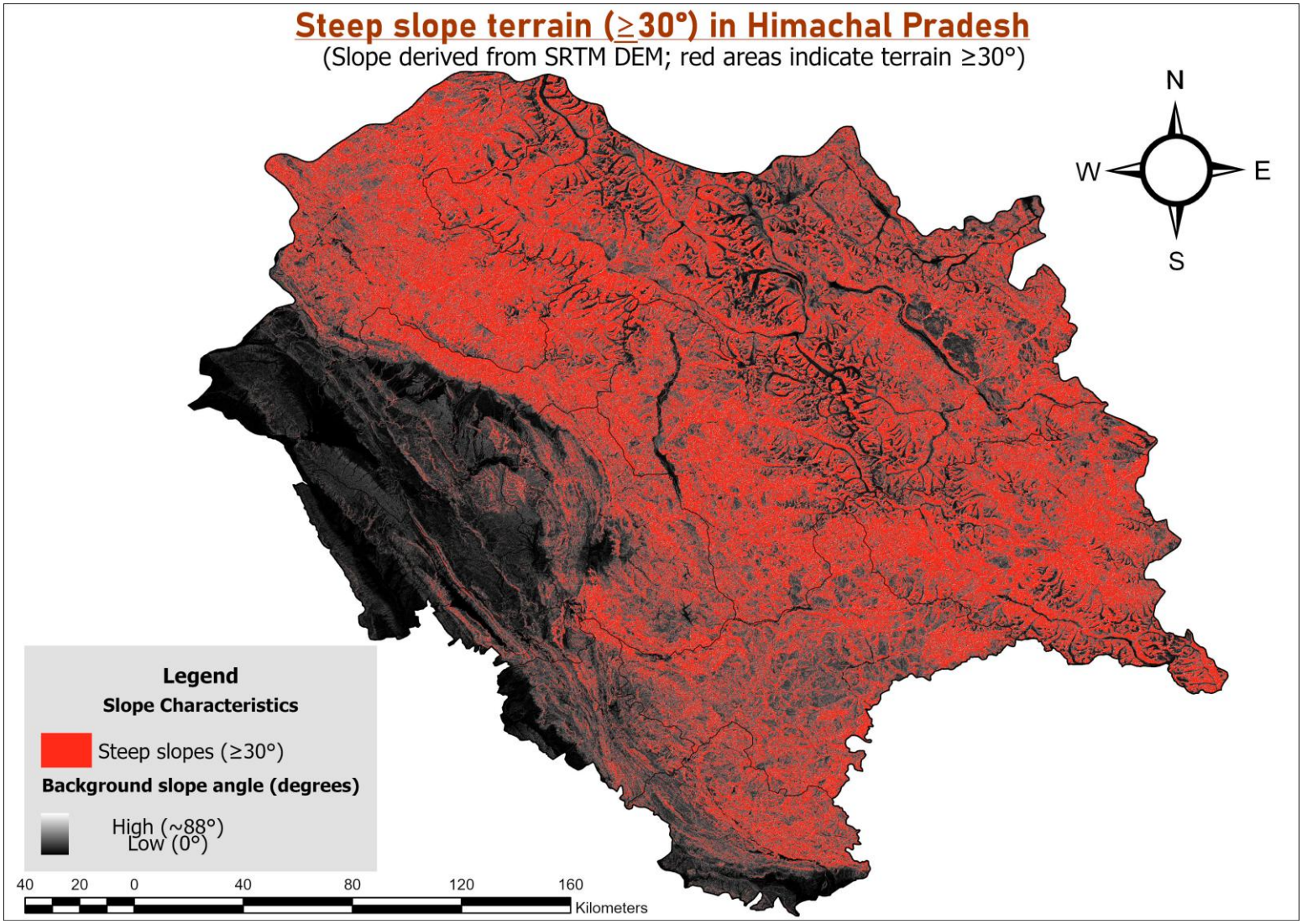


Fig. 2. Steep Slope Terrain ≥ 30°, Himachal Pradesh



2.1 NDVI time-series construction and disturbance detection

Vegetation dynamics were analysed using annual Normalised Difference Vegetation Index (NDVI) composites derived from the Landsat archive for the period 2000–2024. Landsat imagery was chosen for its long-term continuity, consistent spatial resolution, and suitability for regional-scale interannual vegetation monitoring (Roy et al., 2014). To minimise short-term noise from clouds, atmospheric variation, and seasonal phenology, NDVI values were aggregated into annual composites, creating a temporally consistent series appropriate for long-term disturbance analysis (Kennedy et al., 2010; Forkel et al., 2013).

Vegetation disturbance and recovery were detected using the LandTrendr temporal segmentation algorithm, which models NDVI trajectories as a series of connected linear segments to identify breakpoints corresponding to vegetation change (Kennedy et al., 2010). From the LandTrendr outputs, three metrics were extracted: year of strongest disturbance, disturbance magnitude (NDVI loss), and recovery duration. These metrics characterise disturbance timing, severity, and persistence and form the basis for subsequent analyses and results (Figs. 3–5).

2.2 Terrain conditioning and NDVI trend analysis

To isolate landslide-relevant environments, disturbance metrics were conditioned on terrain properties derived from a digital elevation model. Slope served as the primary constraint, retaining disturbance pixels on slopes ≥30° to represent terrain where vegetation loss most strongly affects slope stability. Elevation bands were applied to further stratify disturbance magnitude and recovery duration, recognising the significant altitudinal influence on vegetation and geomorphic processes. This terrain conditioning prevents conflation of low-risk valley disturbances with steep-slope processes and directly supports Research Questions 2 and 4.

Long-term vegetation condition was assessed using the Sen slope estimator, calculated from the full 2000–2024 NDVI time series to map persistent greening or degradation trends (Fig. 6). Sen slope values were sampled from disturbed steep-slope areas and from nearby stable terrain, enabling comparison of trend behaviour and addressing Research Question 3.

2.3 Hotspot synthesis, rainfall context, and sensitivity testing

A coupled landslide-stress hotspot layer was generated by intersecting three conditions: steep terrain (slope ≥30°), significant vegetation disturbance and/or delayed recovery, and negative long-term NDVI trends. Pixels meeting all criteria were classified as stress hotspots. This classification does not represent landslide occurrence or probability but instead delineates zones of compounded environmental stress where terrain and vegetation indicators overlap. The intersection-based method is intentionally conservative, minimising false positives by requiring multiple independent stress signals to coincide.

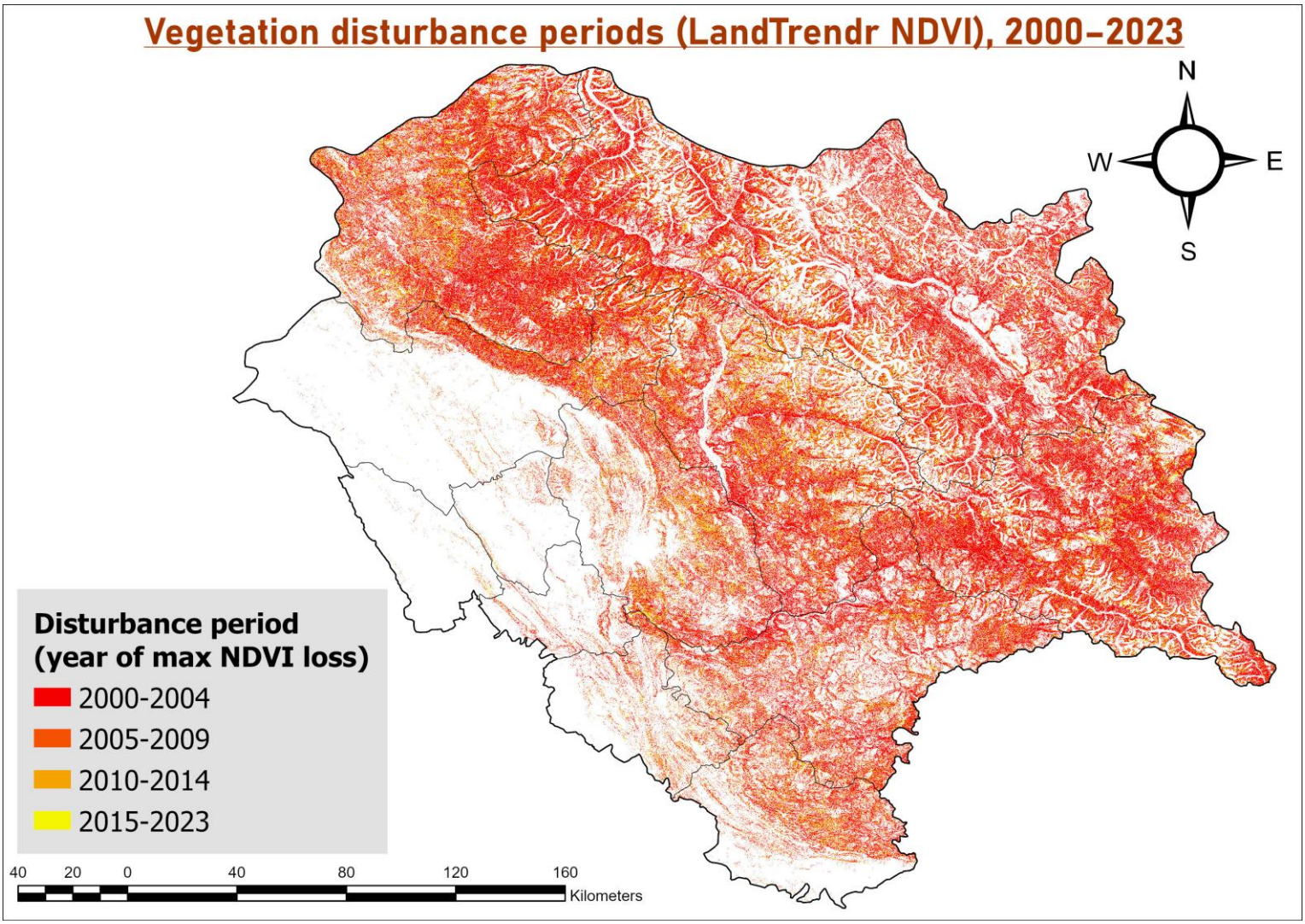
Hotspot areas were aggregated by administrative district boundaries to produce a governance-relevant summary, with total hotspot area (km²) calculated for each district to generate a ranked prioritisation table (Table T1). Monsoon rainfall anomalies derived from CHIRPS data were analysed to provide broader climatic context (Fig. 7) and were used solely for interpretation rather than classification. Finally, a sensitivity analysis was conducted by repeating key area calculations using a ≥25° slope threshold, with differences in hotspot extent and district rankings quantified to assess result robustness (Table T2) without altering the primary analytical framework.

3. Results

3.1 Vegetation disturbance timing

The map showing the spatial distribution of the year of the strongest vegetation disturbance indicates a large but time-differentiated change across Himachal Pradesh from 2000 to 2024 (Fig. 3). Disturbance years differ over different areas of the state; rather, certain locations have disturbance years indicating that the change was episodic rather than continuous. Areas with very steep slopes, especially in the mid and high altitude regions, and the timing of disturbances show that significant vegetation changes happened in those areas during certain times rather than being gradual decline over time. Some districts show disturbance periods that overlap with each other, which indicates the operation of regional scale factors such as land use change, infrastructure development, or climatic stress that influence over multi-year timescales. The timing of disturbances gives the necessary temporal context, but it still does not reveal the intensity or duration of the disturbances.

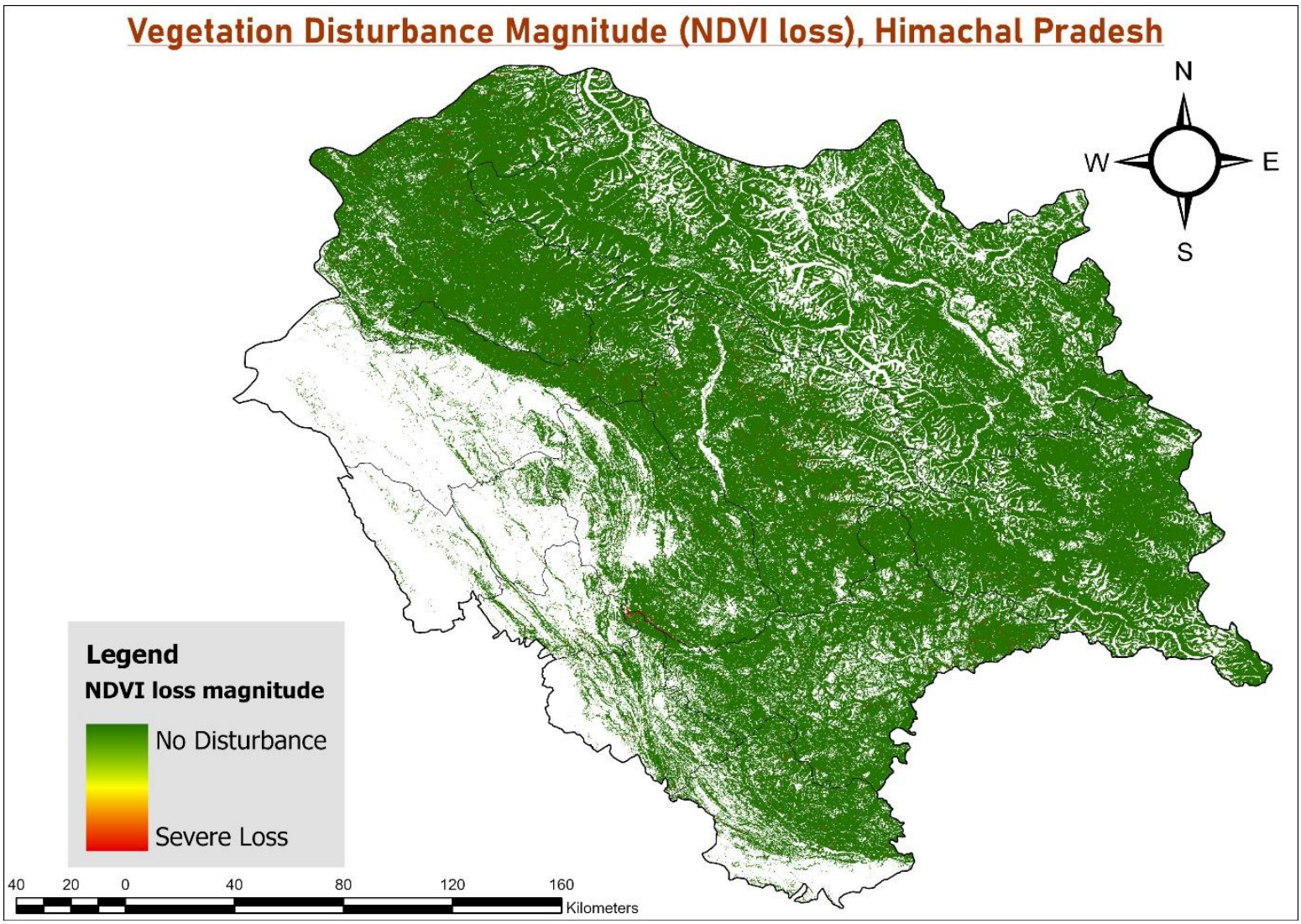
Fig. 3. Year of vegetation disturbance (LandTrendr NDVI)



3.2 Vegetation disturbance magnitude

The magnitude of vegetation disturbance that is represented by NDVI loss, has a very strong spatial variability in the whole study area (Fig. 4). The patches of high magnitude disturbance are lying close to each other; they are revealing clear concentrations of strong dips in NDVI over very steep slopes and also in landscapes where forest is the main land cover. The lower NDVI values in these areas reflect not the short term rise and fall of canopies but the actual loss of trees. While many of the high magnitude zones overlap with the disturbance timing patterns shown in Fig. 3, their area is more limited, which would imply that not every disturbance event leads to such severe degradation. The lower magnitude disturbances are located very widely and are attending in a variety of terrain conditions, thereby indicating a gradient of disturbance intensity rather than a clear line drawn between the areas affected and the ones not affected.

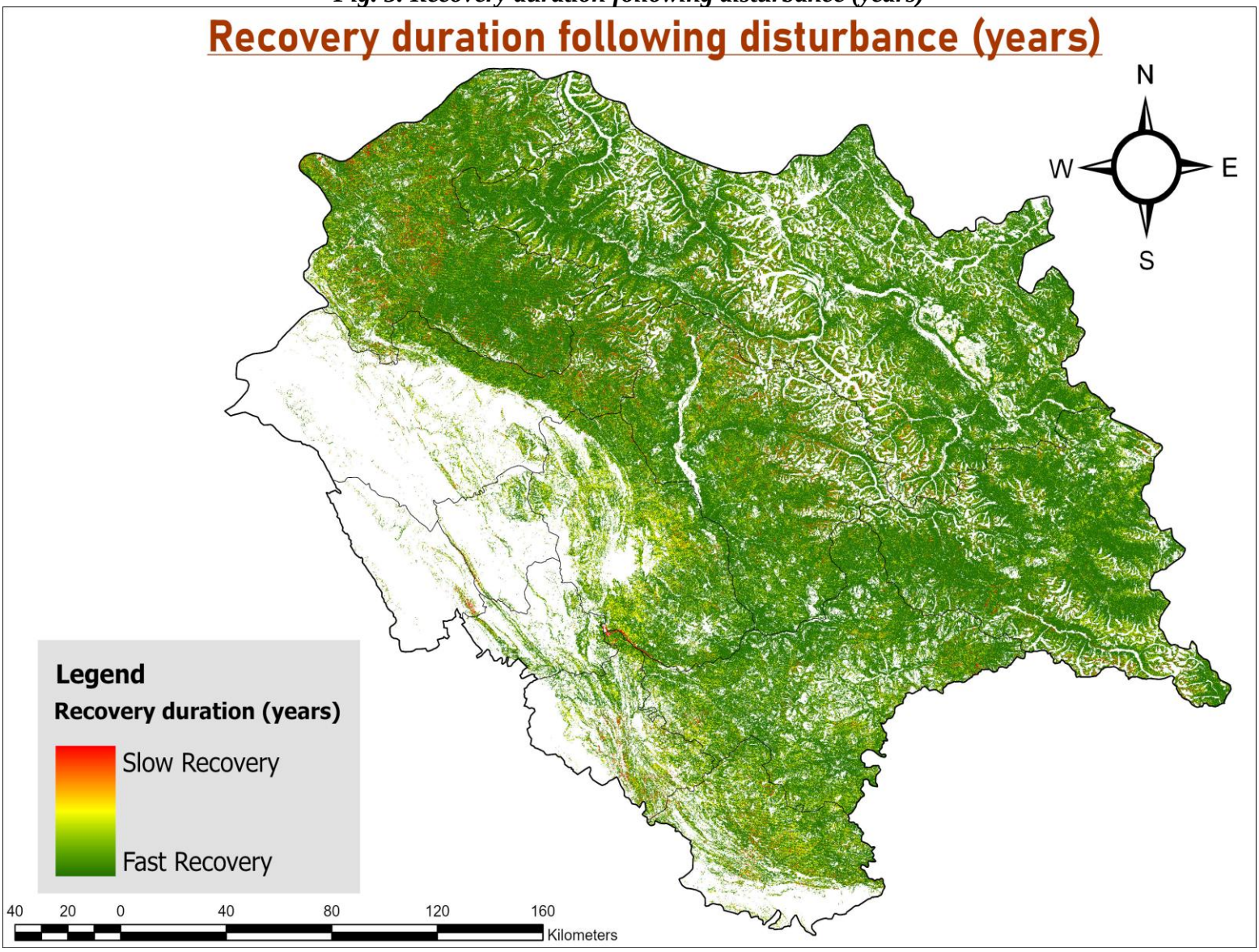
Fig. 4. Vegetation disturbance magnitude (NDVI loss)



3.3 Recovery duration following disturbance

The recovery duration patterns exhibit distinct geographical differences in the dynamics of the post-disturbance vegetation across the whole province of Himachal Pradesh (Fig. 5). The areas that suffered the most underwent recovery taking a long time, as the NDVI (Normalized Difference Vegetation Index) of these areas was below the normal for several years after the event. Delayed recovery is particularly obvious on steep slopes and at high altitudes where climatic factors, very thin top soil, or even human activity might be the reasons for the limited regrowth. On the other hand, the areas at low elevation tend to recover rapidly thus showing greater ecological resilience. Moreover, areas with long recovery durations frequently overlap with high disturbance magnitude zones which implies that the severity of the disturbance has a great impact on the recovery paths.

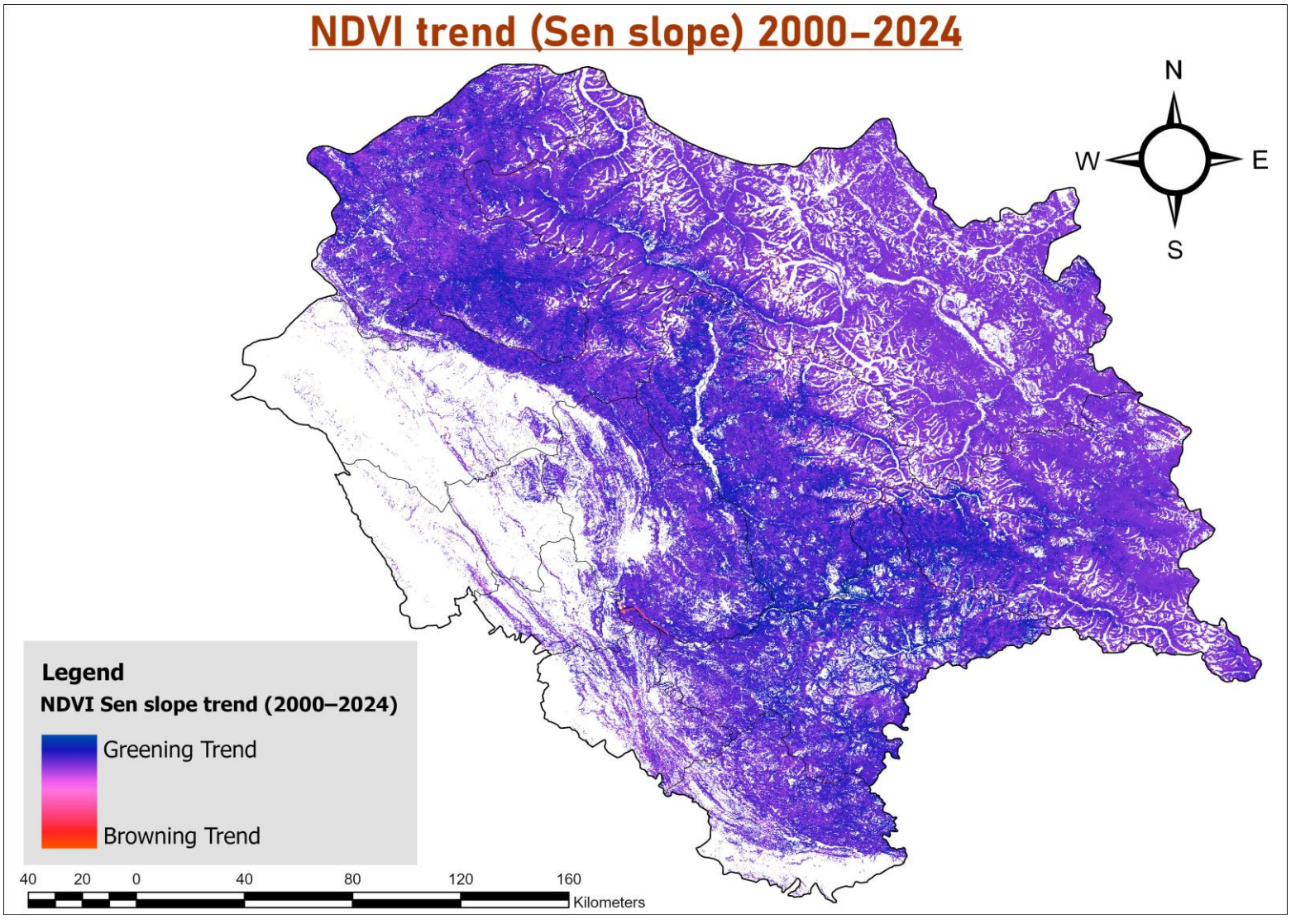
Fig. 5. Recovery duration following disturbance (years)



3.4 Long-term NDVI trend analysis

The NDVI trends over a long period, assessed by the Sen slope estimator, present a large area of negative shifts in vegetation for the state of Himachal Pradesh (Fig. 6). The most significant declines are found in areas that were previously disturbed and therefore are still not able to recover. Positive or nearly neutral trends are limited to a few locations, which are typically outside the steepest slopes or where the disturbance is faint. Overall, the NDVI trend map suggests that the degradation of vegetation is highly concentrated in specific terrain and disturbance settings.

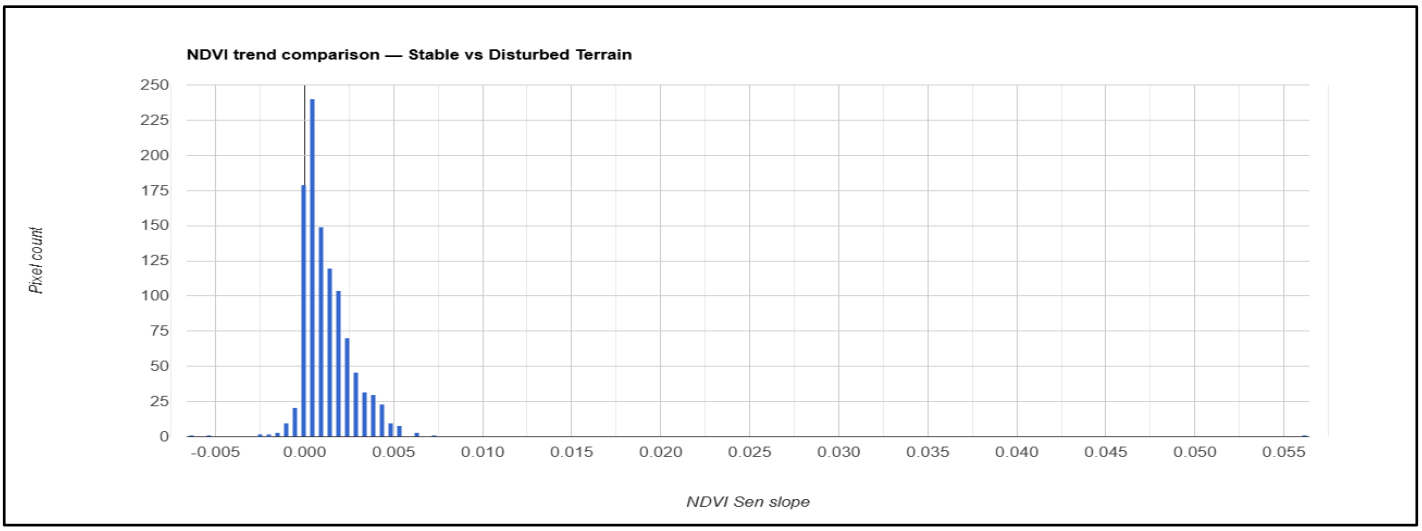
Fig. 6. NDVI Sen-slope trend (2000–2024)



3.5 NDVI trend comparison: disturbed vs. stable terrain

The comparison of NDVI Sen slope distributions for disturbed and stable areas shows a difference in the behaviour of the vegetation trend (Chart). The areas of disturbed steep slopes have distributions that are pushed towards Sen slope values that are more negative, which means that there is stronger and more persistent degradation. The stable area has a narrower distribution that is mainly around zero, indicating that the vegetation conditions are relatively stable. While there is some overlap, the pattern overall points to consistently worse vegetation trajectories in the disturbed areas.

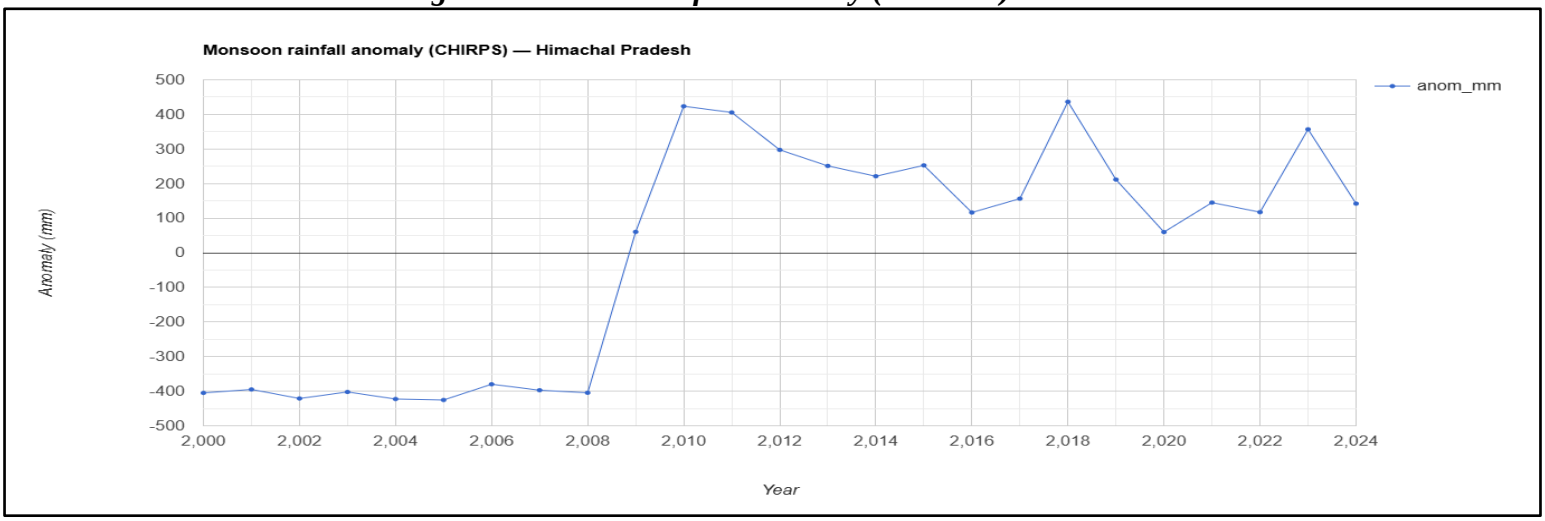
NDVI trend comparison - Stable vs. disturbed terrain (Chart)



3.6 Monsoon rainfall anomaly context

Monsoon rainfall anomalies derived from CHIRPS data display pronounced interannual variability across Himachal Pradesh from 2000 to 2024 (Fig. 7). Several anomalously wet or dry years coincide with the vegetation disturbance clusters identified earlier, although this relationship is not spatially consistent. Rainfall anomalies are included as contextual information only and are not interpreted as direct drivers of vegetation disturbance or hotspot formation.

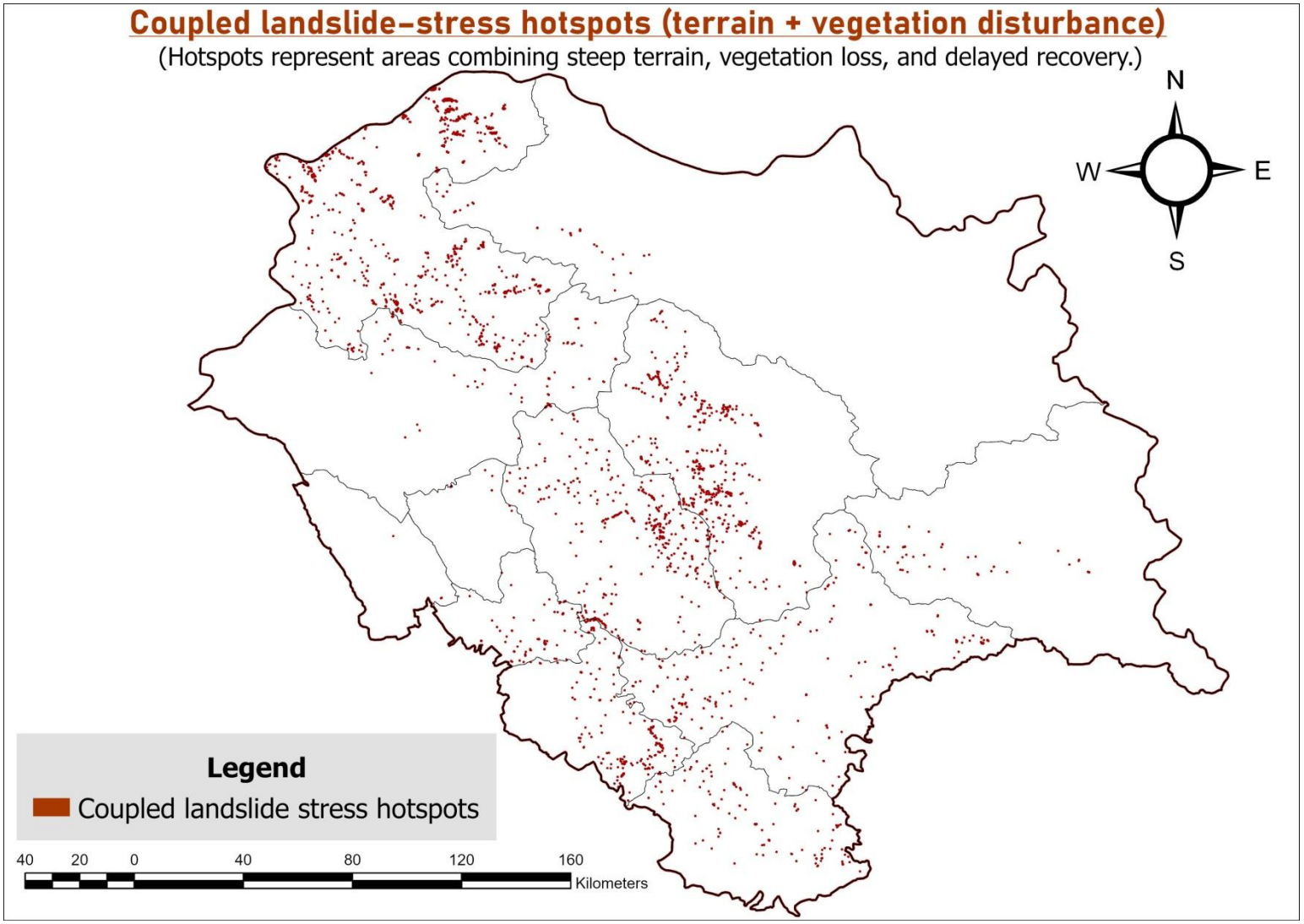
Fig. 7. Monsoon Rainfall Anomaly (CHIRPS)



3.7 Coupled landslide-stress hotspots

The merged landslide stress hotspot map amalgamates steep slopes, vegetation disturbance, slow recovery, and long term NDVI degradation into a singular spatial representation (Fig. 8). The stress hotspots are distributed over the terrain creating zones of overlapping environmental stresses. They mainly exist on steep slopes and often coincides with areas afflicted by high disturbance magnitude and long recovery, especially in mid- to high-elevation regions. The hotspot layer does not show where landslides have occurred but rather indicates places where several stress indicators intersect.

Fig. 8. Coupled landslide-stress hotspots (terrain + vegetation disturbance).



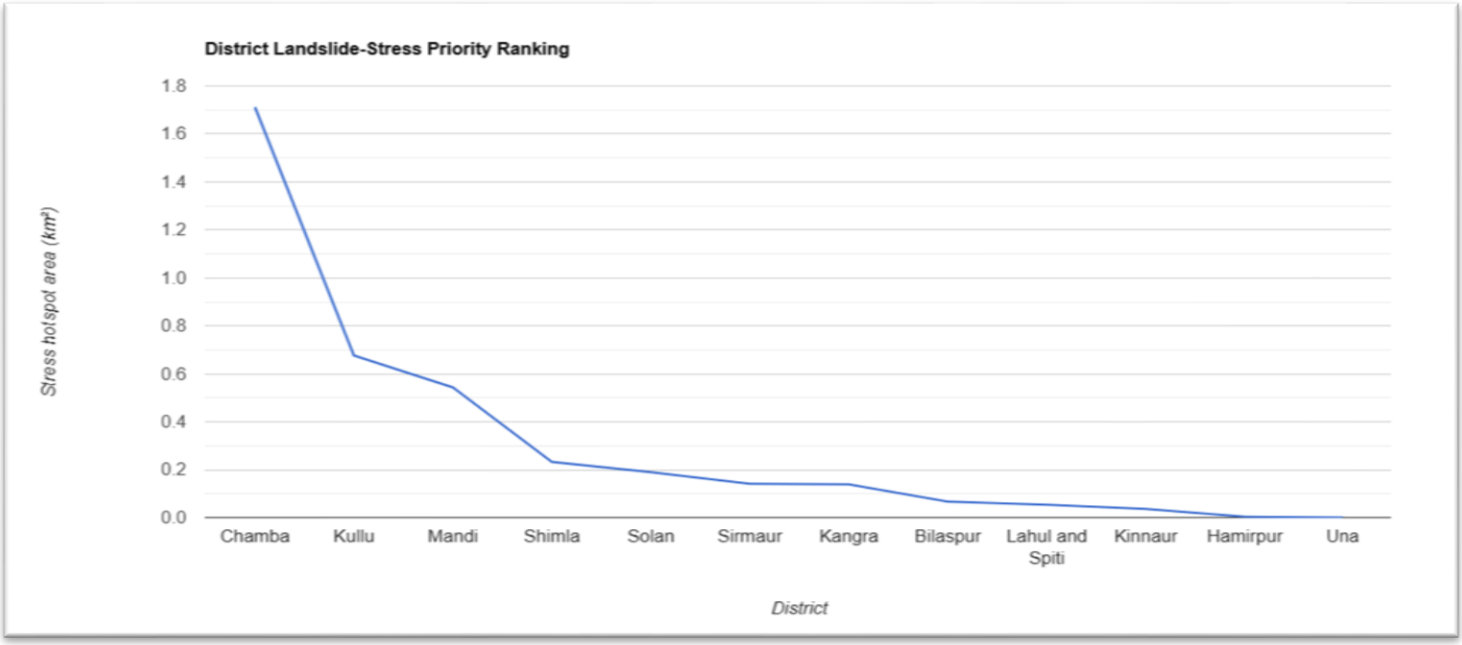
3.8 District-level prioritisation

The stress hot spot areas combined by district exhibit a clear disparity in the total environmental stress (Table T1). A number of districts have a disproportionately large share of hotspot area while the rest are practically stress-free. Such a distribution makes certain districts emerge as targets for monitoring.

T1. District hotspot / stress area ranking

S.No.	District Name	Stress (km2)
1.	Chamba	1.711658939
2.	Kullu	0.680044402
3.	Mandi	0.543758834
4.	Shimla	0.233220618
5.	Solan	0.190585458
6.	Sirmaur	0.141882414
7.	Kangra	0.139692829
8.	Bilaspur	0.067457224
9.	Lahul and Spiti	0.054360268
10.	Kinnaur	0.03669491
11.	Hamirpur	0.003050668
12.	Una	0

District Landslide-Stress Priority Ranking (chart)



3.9 Sensitivity to slope threshold selection

Sensitivity analysis comparing slope thresholds of ≥25° and ≥30° reveals that total hotspot area grows with the lower threshold, yet relative spatial patterns and district rankings stay consistent. This robustness to slope threshold choice strengthens confidence in the findings.

T2. Sensitivity (25° vs 30° slope threshold)

Threshold	Hotspot area (km²)
≥30°	0.000005206349956 km²
≥25°	0.000005204811637 km²
Difference	−0.000000001538319 km²
% Change	−0.0295 %

The very similar numbers and the slight reversal most probably are the result of raster discretisation/rounding and the extremely small size of the hotspot; the sensitivity test shows that there is almost no difference in the total hotspot area when the two thresholds are applied.

4. Discussion

Vegetation disturbance as a driver of landslide-related stress

Himachal Pradesh, in terms of vegetation disturbance, is a region where the disturbances are neither temporal uniformity nor random in terms of location; rather, they are distinctly clustered in the steep Himalayan areas. This characteristic reinforces the view of the vegetation disturbance being a landscape-wide environmental stressor rather than an isolated ecological process. Vegetation gives support to the slopes by root reinforcement, better soil cohesion, and reducing surface runoff. Thus, the gradual loss of vegetation on steep slopes will have geomorphic consequences, even though there are no direct landslide records. The study's results on the occurrence of high-impact disturbances and slow recovery quite match the models linking the decline of vegetation to increased slope instability in mountain landscapes (Chen et al., 2019). This research does not establish a direct connection between particular landslide occurrences and the disturbance of vegetation. Instead, it points out the areas where the environmental conditions favoring slope instability are intensifying with time, thus providing a cautious, intermediate stance that connects ecological monitoring with landslide susceptibility evaluation.

Disturbance severity, recovery delay, and long-term degradation

One of the major advantages of the study is its comprehensive approach in combining disturbance timing, magnitude, recovery duration, and long-term NDVI trends. Disturbance events may happen in different areas, but it is only in some regions that great NDVI falls and long recovery periods coincide. This emphasizes disturbance severity as a major factor controlling long-term vegetation patterns. The slow recovery process on steep, high-elevation areas is probably due to the harsh climate, poor soils, and human impacts that are still ongoing, which parallels the results of the previous time-series studies (Kennedy et al., 2010; Forkel et al., 2013). The Negative Sen-slope trends in the areas affected by disturbances indicate that the decrease of vegetation is not a single event but rather a trend of continuous decline during the 2000–2024 period. This is significant because the continued negative shifts indicate the presence of a cumulative environmental stress rather than just the case of brief disturbances followed by the recovery.

Terrain controls and spatial coherence of stress

Slope and elevation analysis discloses the uneven patterns of disturbance magnitude, recovery time, and NDVI drop across the different soil types. The steep slopes (30° and higher) and the mid-to-high elevation areas are the ones that bear the heaviest burden of the strong and still-existing disturbance signals, which demonstrates the important role of the type of the terrain in the vegetation's ability to recover or die. The factors that probably contribute to the amplification of the disturbance impact on these slopes are: thin soils, increased erosion, and gravity, which in turn lead to slow recovery and a higher risk of degradation, and these patterns are evident across the slopes regardless of the threshold. The combined landslide-stress hotspot map incorporates steep terrain, vegetation disturbance, delayed recovery, and long-term decline of NDVI into a single clear spatial indicator. Some distinct hotspot clusters are formed, indicating that the environmental stress clusters are positioned in areas rather than being scattered, and this is because of the interactions among topography, vegetation dynamics, and climate variability which Pacheco-Quevedo et al., 2023 support. The conservative, intersection-based definition of hotspots further strengthens a meticulous, evidence-based methodology.

District-level implications and rainfall context

The aggregation of data at the district level shows a great difference in the distribution of environmental stress, where only a few districts have a disproportionately large number of hotspot areas. This situation is similar to the regional pattern of landslide risk in the northwestern Himalaya (Gautam et al., 2022; Singh et al., 2024), although in this case, it is caused by the gradual changes in vegetation over time rather than by fixed factors. The anomalies in monsoon rainfall are a good indicator of the hydrological and climatic situation, as many periods of disturbances coincide with the years that were abnormally wet or dry. Yet, the rainfall was intentionally left out of the hotspot classification because the purpose was to prevent mixing the short-term triggering effects with the long-term accumulation of environmental stress (Zhu et al., 2025).

Methodological perspective

One significant advantage of this research is the application of a protracted and uniform Landsat time series, which allows for dependable observation of the vegetation disturbance and recovery throughout more than 20 years (Roy et al., 2014). Considering vegetation

disturbance as a surrogate for environmental stress, instead of a direct sign of landslide occurrence, shows the study's cautious and monitoring-friendly approach.

5. Limitations and Future Work

One of the main drawbacks of this study is the absence of a comprehensive landslide inventory. In the absence of spatiotemporal records of landslide occurrences, it is not possible to directly validate the disturbance of vegetation with the real events. Thus, the study does not make predictions or conduct vulnerability models; instead, it interprets the results as signs of landscape scale environmental stress, which is in line with monitoring oriented frameworks.

The evaluation depends on surface indicators obtained through remote sensing visual methods, mainly focusing on vegetation dynamics and terrain geometry. Subsurface elements such as lithology, soil composition, and structural geology, which are crucial to slope stability, were excluded. Although this keeps the approach simple, it limits the ability to differentiate between geomorphological and vegetation-related stress signals. The Landsat data archive of images, although not having the best spatial resolution, provides great temporal coverage, thus, it is still very useful for identifying landslide scars or narrow road cuts.

Although there is a possibility to use optical or LiDAR datasets that have higher resolution, they usually do not have the same temporal depth. Rainfall variability was taken into account only as background information. In the future, the work could fuse rainfall intensity metrics, soil moisture indicators, or hydrological models to conduct examinations of interactions produced by stress. Other extensions include the fusion of multi-sensor data, the explicit evaluation of infrastructure growth per area, and the integration with landslide susceptibility models when the necessary training data is available.

6. Conclusion

A multi-temporal Landsat framework was adopted in this study to trace the path of vegetation disturbance, recovery, and eventual degradation in the steep terrain of Himalayas, Himachal Pradesh from the year 2000 to 2024. The application of disturbance detection, terrain stratification, and trend analysis of NDVI led to the discovery of environmental stress at the landscape level associated with the landslide-prone locations. The disturbance to the vegetation cover occurred almost everywhere but the worst of the situation was on the steep slopes and in the particular elevation zones. The recovery of these areas has had a lot of differences, as some have been showing continuous decline in NDVI which means they have been experiencing cumulative stress rather than just transient.

The combination of terrain with district level analysis has resulted in the identification of areas where disturbance, slow recovery, and degradation coincide i.e. the areas are marked with severe environmental stress. The study has not resorted to predictive modeling but rather kept a conservative stress monitoring perspective and it demonstrates the importance of long-term Earth Observation time series in assessing environmental stress and supporting landscape monitoring in mountainous areas.

GEE CODE URL: https://code.earthengine.google.com/02beb2d860d96f433856d23851a0be4d

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